NAME: **AGBEYEBIAWO DENNIS** DATE: MAY 14, 2022.

INDEX NUMBER: **3614318**

PROGRAMME: **BSC MARINE ENGINEERING**

COURSE: **SHIP MACHINERY SYSTEM DESIGN (MME 473).**

MID-SEMESTER ASSIGNMENT

CASE HARDENING

1. Case hardening is a process that strengthens a metal surface by creating a thin coating of a harder metal alloy over the original metal's softer core. Case hardening is a material processing technique used to increase the hardness of a metal's outer surface. Molecules and atoms are packed tightly into crystalline structures in solid metals. The outer layer's strength and hardness increase significantly when carbon diffuses into the crystalline structure from carbon-bearing matter on the surface of low carbon steel or alloy steel. The case hardening process tries to solve the brittle mass formation problem.

   The **advantages of case hardening** will be explained further down:

   i. Enhances the durability and cost-effectiveness of steel parts.

      The components' wear resistance and fatigue life are considerably improved by the mechanical strength and surface hardness created by case hardening and the retention of the softcore. The ability to absorb the energy released by impact loads is improved by retaining a softer core, which leads to longevity and cost savings.

   ii. Enables the use of steel with superior machinability for heavy-duty applications.

Alloys used for heavy-duty applications have low machinability because they must be harder and stronger. Case hardening process enables the use of low carbon steel with precision machinability for weapons and firearms applications and other similar heavy-duty uses that require mechanical strength, fine finishes, and accurate geometry. Subsequent case hardening after machining provides superior wear resistance and hardness on the surface of the precision machined parts.

iii. Enhances the performance of taper roller bearings.

Case hardening leaves a softer core that is advantageous for accommodating a misalignment condition common in taper roller bearings. A strong surface layer with a soft inner core ensures fracture resistance under an extreme operating environment that includes misalignment, vibration, and severe shock loads

iv. Enables formation of residual compressive stresses in gear teeth

Case hardening on gears enables the formation of residual compressive stresses both inside the case depth and on the surface, which avoids crack formation and neutralizes bending stress. Advanced induction hardening methods have enabled accurate case hardening of heavy-duty spiral bevel gears, pinion, hypoid gears, and helical gears of complex geometry.

v. Weldability of steel is improved by case hardening

Case hardening improves the weldability of steel, which is important for certain engineering applications.

vi. Minimizes Distortion

Case hardening produces a low-cost, wear-resistant load-bearing surface with minimal distortion.

vii. Produces improved fatigue resistance

There are different processes of case hardening.

I. Carburizing

Carburizing is one of the types of the heat treatment process and it is also known as carburization. In the carburizing technique, the metal is heated in the presence of charcoal or carbon monoxide. Carburizing is to make the metal harder by heating it in a very short interval of time. In the process of carburizing, various hardened metals are used like carbon monoxide, barium carbonate, sodium cyanide, etc.

After completing carburizing, you may see the changes in size and shape of the original metal. But these changes are very small compared to other heat treatment processes.

II. Nitriding

Nitriding is one of the heat treatment processes in which nitrogen spread over the surface of the metal to form its layer on the surface of the metal. Nitriding is one of the parts of the case hardening. It is used mostly in the case of steels that contain low carbon content. There are some expectations to the previous like titanium and molybdenum. Nitriding has a very larger number of applications like it is used in making gears, injectors, screws, camshafts, and various tools.

Nitriding is divided into three main processes :

a. Plasma nitriding

Plasma nitriding is mostly used in industrial applications. Plasma nitriding is also termed ion nitriding or glow-discharge nitriding. In plasma nitriding, an effective electric field is generated which results in the ionization of the gas molecules. In the entire process of plasma nitriding, nitrogen is used. It is the

perfect process for treating steel. In the plasma nitriding process, along with the nitrogen certain amount of hydrogen and argon are also used.

b. Gas nitriding

In gas nitriding, ammonia is used. During gas nitriding, ammonia comes in contact with heated metal and this ammonia separates into hydrogen and nitrogen. The nitrogen formed then spread on the surface of the metal forming nitride layer.

c. Salt bath nitriding

In the salt bath nitriding, the cyanide salt plays an important role. The temperature is between the 550-570º C. The most beneficial part of the salt bath trading is that you get the diffusion of the cyanide salt over a larger surface area than any other process in the same time interval. This process is done very quickly and requires only 4 hours. Along with the quick process, there are other complications involved in this process.

III. Carbonitriding

Carbonitriding is the type of the case hardening process in which the hardness of the metal surface is increased and also any type of deformation is also reduced during this process. As the name suggests, in carbonitriding, the carbon and nitrogen atoms spread over the metal surface and result in increasing the hardness of the metal. The carbonitriding is carried out above 800º C. The layer that forms on the workpiece as a result of the carbonitriding is quite hard. This layer is of average thickness ranging from

0.07 mm to 0.5 mm. The maximum thickness of the layer that can be formed is 0.75 mm.

IV. Cyaniding

Cyaniding is a case hardening process in which the metal is heated in the temperature range of 871 to 954º C. Here, the metal part is heated in the presence of sodium cyanide. Cyaniding results in the formation of a layer of thickness ranging from 0.25 mm to 0.75 mm. This process is typically applicable for low-carbon steels. Also, unlike other processes, it requires only 20 to 30 minutes. Its application is found on metallic parts such as gears, screws, small nuts, etc. The disadvantage of this process is that here cyanide salt is used and this cyanide salt is very poisonous and this hazardous to our life. On the other side, its advantage is that it is an efficient and fast process.

V. Ferro Nitrocarburizing

Ferric nitrocarburizing is one of the case hardening processes in which the carbon and nitrogen are spread out on the ferrous metals. . Ferric nitrocarburizing is carried out at the sub-critical temperatures during the process of the salt bath. This process is practiced mostly in the temperature range of 525ºC to 625º C. During such temperatures, the steels, and other alloys are in the ferritic phase.

Examples of components made from case hardening are :

I. Gears
II. Fasteners
III. Camshaft
IV. Rods

V. Pins

Strength and stiffness are two physical properties of a material. One major difference between them is that stiffness is the ability of an object to withstand stress without breaking. On the other hand, strength is the ability of an object to resist deformation when stress is applied. Strength and stiffness are two physical properties of a material. One major difference between them is that stiffness is the ability of an object to withstand stress without breaking. On the other hand, strength is the ability of an object to resist deformation when stress is applied. A strong material with less stiffness will break if the exerted force exceeds its tensile strength. The strong material does not change its form. If the exerted force exceeds its strength, it simply breaks, completely losing its original shape. However, stiff material with less strength will deflect. The deflection helps it to accommodate the force exerted on it. Therefore, it can return to its original shape once the force is removed.

A ship structure is sufficiently strong if it can support all the fixed loads of its machinery, fittings and equipment, and all the cargo loads, and if it can withstand without permanent change of shape, cracking, and fracture all the hydrostatic and hydrodynamic loads which can be imposed upon it in calm water or in any kind of seaway in which it is supposed to operate. It is sufficiently stiff if, under any of these service loads, it does not flex or deform unduly to interfere with the alignment of operation of machinery or with any other function, major or minor.

A bronze screw propeller of a high-powered ship, having blades that are too thin, may deform so much when generating full thrust that its shape and propulsive characteristics are changed.

If the blades are permanently bent under emergency maneuvering conditions, the propeller is not strong enough. Vibratory motion can build up to unacceptable peaks if local portions of the structure are in resonance with the periodically applied forces. In consequence, rudder plating and shell plating at the stern must be stiffened to control panting or pulsating deflections from propeller forces. The shell at the bow is also subject to panting from wave-impact loadings; this requires extra stiffeners, called panting stringers.

2. A crankshaft is the heart of an internal combustion engine, and it cannot function without it. It uses unbalanced masses called crank webs to allow the pistons to constantly revolve inside the cylinder. Its solid geometry is complex.

The crankshaft is made up of three primary components: the crank pin, the crank web, and the shaft. The connecting rod's big end is attached to the crankpin, and the crank web connects the crankpin to the shaft part, which is rotated by the main bearings and sends power to an external source through a belt, gear, or chain drive. Through the connecting rod, the load of the gas forces inside the combustion chamber is transferred and spread over each crankpin. Every crank web is subjected to bending moment and twisting moment. There is a flywheel attached at the end of the crankshaft to bring uniformity in torque of a four-stroke I.C. engine by storing energy during the power stroke and releasing the same during the other three strokes.

The crankshaft should have sufficiently high strength to sustain the gas force acting vertically downwards during the expansion stroke. The strength of the crankshaft should be high enough to avoid bending failure. Hence, the crankshaft greatly affects the life and reliability of an I.C. engine.

The following stresses directly affect the crankshaft of an I.C. engine:

i. Bending Stress: - The gas force generated by the burning of the air-fuel mixture in the combustion chamber, above the piston head, forces the piston downwards. This force is transmitted to the crankpin bearing. It is a bending load, causing corresponding bending stress.

ii. Torsional Shear Stress: - Crankshaft is a rotating component, and it runs at high speeds. Any rotating mass generates centrifugal force greater the engine speed, the greater the centrifugal force. Connecting the rod also generates centrifugal force. Due to this, both, bending and torsional shear stress are developed in the crankshaft. These areas